AI-Driven Aviation Analytics: Revenue & Route Optimisation

EXECUTIVE PROJECT BRIEF

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A strategic initiative to transition airline operations from static models to dynamic, data-driven decision-making.

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1. The Business Problem

Airlines operate in a highly volatile industry characterised by thin profit margins, fluctuating fuel costs, and intense competition. Currently, the airline's approach to pricing, route management, and fleet deployment is largely static and heavily reliant on generalised historical averages rather than granular, real-time data.

This "one-size-fits-all" operational model has led to three critical inefficiencies:

Missed Yield Opportunities

Ticket pricing remains relatively flat throughout the day, failing to capture consumer surplus during peak hourly and daily booking windows.

Cabin Misalignment

There is a lack of clarity regarding the true revenue drivers within the aircraft. While Economy seating dictates volume, the disproportionate revenue impact of Business and Comfort classes is not being fully leveraged in marketing or upgrade campaigns.

Inefficient Fleet Utilisation

High-capacity aircraft and expensive crew resources are being deployed on underperforming routes with consistently low load factors, resulting in unnecessary operational waste.

Without a data-driven intervention, the airline will continue to leave significant revenue on the table while absorbing the costs of under-optimised flight paths.

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2. The Project Objective

The primary objective of this project is to leverage advanced data analytics—accelerated by AI-driven SQL validation and modelling—to transition the airline from a static operational model to a dynamic, highly responsive enterprise.

By analysing the comprehensive dataset of flights, ticketing, and booking references, this project aims to deliver actionable business intelligence across the following pillars:

01

Identify Booking Behaviours

Map the daily and hourly trends of customer bookings to establish a foundation for "Time-Based Dynamic Pricing" and targeted flash sales.

02

Analyse Revenue Distribution

Quantify the exact financial impact of different fare classes (Economy vs. Business) and specific aircraft models to drive future decisions on cabin retrofitting and targeted upsell strategies.

03

Perform Route Optimisation

Isolate the top 5 highest-yielding routes to protect market share, while identifying the bottom 5 underperforming routes for immediate strategic review, marketing intervention, or potential cancellation.

3. Expected Outcomes

By integrating relational database management (PostgreSQL) with dynamic executive dashboards (Excel), this project will equip stakeholders with the verifiable, interactive insights required to:

Increase Net Profit Margins

Without necessarily increasing flight volume.

Reduce Operational Overhead

By reallocating resources away from dead-weight routes.

Enhance Market Competitiveness

Through agile, data-backed decision-making.

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